

Application Developer (Developer 3)

CAM/AUTO | Mumbai | Full time or 5-month contract | Job description, 29 September 2026

Company	CAM/AUTO, the in-house CAM software of a precision CNC machine shop
Location	Mumbai. Hybrid acceptable: at least two days a week on site for the shop tablet page, dry runs and learning reviews
Type	Full time or a 5-month contract, starting about 4 weeks after the geometry lead and platform engineer. Completion bonus at the week-21 gate; conversion to a phase-two role for a full-time hire
Reports to	Founder (product owner). Day-to-day design direction from the geometry lead; infrastructure from the platform engineer; every learned value approved by the process owner
Compensation	INR 18 to 28 lakh a year (Mumbai 2026 band, to be confirmed), pro rata for a contract, plus a completion bonus tied to the week-21 first-article gate

The role in one paragraph

You will build the parts of CAM/AUTO that the shop floor touches and that make the app learn. The geometry lead makes the engine exact and the platform engineer makes delivery safe; you build the shop tablet page, the database-backed history, the Siemens post validation, adaptive roughing under the lead's design, and the first version of the learning module: models that propose cutting speeds, feeds and size corrections from measured first articles, clamped, replayed on the whole part library, and approved by the process owner before use. This role exists only in the full (3-developer) scope; it is the role that turns collected data into a product that improves with every part.

What you will do

Weeks 4 to 8: join and ship the first screens

- Learn the app from its modules and tests; ship the shop-floor mode of the single-file app: setup card, tool list, measurement entry and tool-wear events on a rugged tablet, writing to the platform engineer's on-site service.
- Per-part history page: every program by hash with its inputs, tools, machine, fixture, operator and measured result.

Weeks 8 to 15: posts and roughing

- Siemens Sinumerik post validation: dry runs on the shop's Siemens mill with the process owner, differences against Haas/Fanuc logged and fixed, every run tied to a program hash.
- Adaptive (trochoidal) roughing and rest machining with a smaller tool, implemented under the geometry lead's design and test contract; the lead reviews every merge.

Weeks 12 to 21: learning module v1

- Data: with the platform engineer, the measurement records joined to program hash, feature, tool, machine, fixture and operator; the data quality checks that reject incomplete records.
- Models, in Python: clamped speeds and feeds per material, tool and machine (regularised regression or small tree models); per-machine, per-tool size-correction offsets from CMM deviations; drawing-reading corrections from operator edits. No deep learning; every output has a clamp the process owner sets.
- The approval loop: offline replay of every learned value on the whole part library through the app's own safety checks; a review screen for the process owner; versioned JSON written back for the planner to read; rollback of any value in one click.
- The one metric: first-article corrections per part, tracked per quarter, on a screen the founder and the board can read.

Weeks 20 to 26: go-live

- Hardening, operator training material with the process owner, and support for the three setter-run jobs.
- Hand over the learning module with a written design note and the tests that guard it.

Technical skills and technologies

Core: must have, tested in the paid task

Area	Technologies	Level	Used for
TypeScript / JavaScript	ES2020+, TypeScript strict mode, ES modules, esbuild, Node 20	Expert	Everything in the app is TypeScript
Front end without a framework	Plain DOM APIs, CSS layout for tablets, offline-tolerant forms, responsive design; three.js awareness	Strong	Shop tablet mode inside the single-file app; the UI has no React or Vue and stays that way
Backend and data	REST clients, Fastify (Node) or FastAPI (Python) endpoints, PostgreSQL queries and migrations, JSON schema	Strong	History page, measurement entry, learned-value storage
Python for tabular ML	Python 3.11, pandas, scikit-learn (linear models with regularisation, decision trees, gradient boosting), cross-validation, calibration	Strong	Learning module v1: speeds, feeds, size corrections
Testing	Vitest or Node test runner, Playwright for the tablet page, pytest for models, golden-file replay tests	Strong	Every learned value replayed; every screen covered

Area	Technologies	Level	Used for
Git and CI	Git, GitHub pull requests, CI that must be green before merge	Working	The platform engineer's pipeline is the gate

Domain: strongly preferred, learnable in the first two months

Area	Technologies	Level
CNC and NC programming	Fanuc/Haas ISO G-code and Siemens Sinumerik (840D) syntax; work offsets, tool tables, cutter compensation; can stand a dry run with the process owner	Working
Toolpath algorithms	Trochoidal / adaptive roughing, rest machining, stepover and engagement control; implements under the geometry lead's design, does not invent geometry	Working
Machining physics	Speeds and feeds, chip load, tool wear, material groups; enough to sanity-check a learned value before the process owner sees it	Working
Metrology data	CMM report formats, gauge readings, drawing tolerances joined to features	Working
AI-assisted development	Claude Code: writes tasks with acceptance tests, reviews generated diffs, catches wrong numbers	Working

Not required

Deep learning frameworks (PyTorch, TensorFlow: v1 is tabular and small); CAD kernel programming (the geometry lead owns it); cloud platforms and Kubernetes (everything runs on site); front-end frameworks.

Experience profile

- 3 to 6 years as a full-stack developer on a product real users depended on, with at least one shipped feature that used a statistical or machine-learned model on tabular data in production.
- Disciplined test writer: can show a feature where the tests were written before the code.
- Comfortable explaining a model's output to a non-programmer and being told it is wrong by someone who runs the machine.
- Likely backgrounds: product start-ups with data features, industrial or IoT software teams, analytics engineering roles that shipped models, not only notebooks.
- Degree in computer science, engineering, statistics or equivalent shipped work.

How we hire

1. 30-minute call with the founder: the plan, the learning module, your questions.
2. **Paid one-day task** on anonymised repository data: given 60 measured first-article records (feature, tool, machine, programmed size, measured size), build a clamped size-correction model with cross-validation and a replay test that rejects any correction outside the clamp; then a one-screen review page for it in plain TypeScript. We pay for the day whether or not we continue.
3. Two-hour review of your task with the founder, the geometry lead and the process owner, at the shop.
4. Reference calls. Offer within a week of step 3.

What you get

- The feature that defines the product's next phase: a CAM system that learns from every part it cuts.
- A shop floor that measures your model's output within a week of shipping it.
- Claude Code on your desk, an on-site workstation (16 cores, 64 GB), the platform already built to run on.
- Completion bonus at the week-21 gate; for a full-time hire, the phase-two role growing the learning module to strategy ranking and setup choice.

Keywords: TypeScript, JavaScript, Node.js, esbuild, vanilla DOM, tablet UI, Playwright, Vitest, Fastify, FastAPI, PostgreSQL, Python, pandas, scikit-learn, regression, gradient boosting, tabular ML, MLOps-lite, model replay, human-in-the-loop, full-stack developer, CNC, G-code, Siemens Sinumerik, adaptive roughing, speeds and feeds, CMM, manufacturing software, on-premises.